

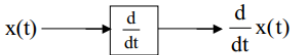
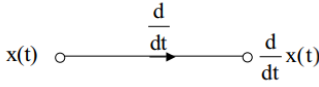
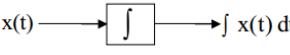
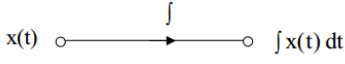
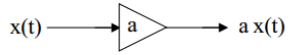
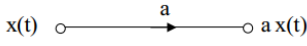
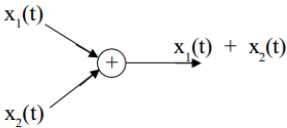
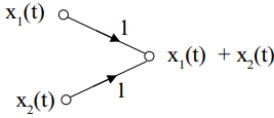
## Day 4 Of Communication System Class

### 2.6.2 Block Diagram and Signal Flow Graph Representation of LTI Continuous Time System

#### Block Diagram

A **block diagram** of a system is a pictorial representation of the functions performed by the system. The block diagram of a system is constructed using the mathematical equation governing the system.

The basic elements of a block diagram are Differentiator, Integrator, Constant Multiplier and Signal Adder. The symbols used for the basic elements and their input-output relation are listed in table 2.2.

| Description                              | Elements of block diagram                                                           | Elements of signal flow graph                                                        |
|------------------------------------------|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| Differentiator                           |  |  |
| Integrator (with zero initial condition) |  |  |
| Constant Multiplier                      |  |  |
| Signal Adder                             |  |  |

### Signal Flow Graph

A **signal flow graph** of a system is a graphical representation of the functions performed by the system. The signal flow graph shows the flow of signals from one point of a system to another and gives the relationship among the signals. The signal flow graph of a system is constructed using the mathematical equation governing the system.

The basic elements of a signal flow graph are nodes and directed branches. Each node represents a signal. The signal at a node is given by the sum of all incoming signals. Each branch has an input node and an output node. The direction of signal flow is marked by an arrow on the branch and the operation performed by the signal is indicated by an operator like integrator/differentiator. When the signal passes from the input node to the output node, it is operated by the operation specified by the branch. The basic operations performed by the branches of a signal flow graph are listed in table 2.2.

### Example 2.7

Construct the block diagram and signal flow graph of the system described by the equation,

$$\frac{d^2 y(t)}{dt^2} + 2 \frac{dy(t)}{dt} + 3 y(t) = 4 \frac{dx(t)}{dt} + 5 x(t)$$

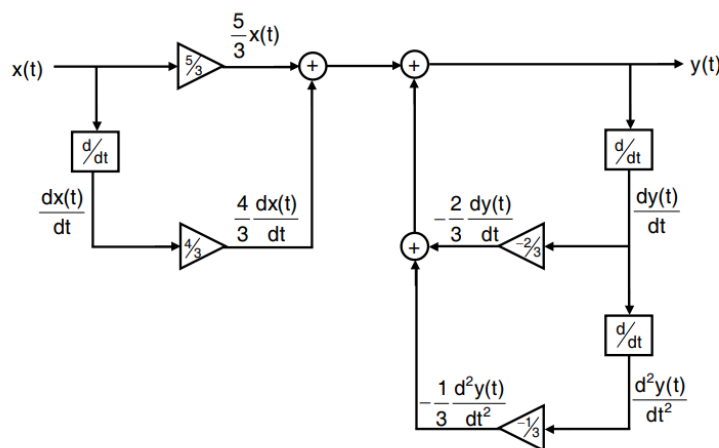


Fig 1 : Block diagram using differentiators.

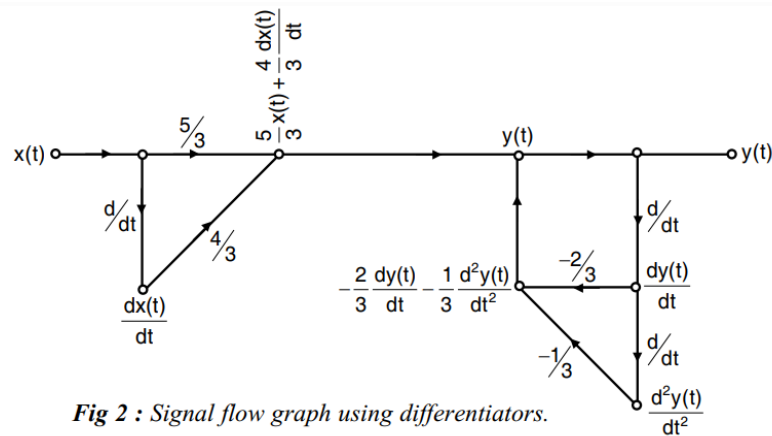


Fig 2 : Signal flow graph using differentiators.

## 2.8 Classification of Continuous Time Systems

The continuous time systems are classified based on their characteristics. Some of the classifications of continuous time systems are,

1. Static and dynamic systems
2. Time invariant and time variant systems
3. Linear and nonlinear systems
4. Causal and noncausal systems
5. Stable and unstable systems
6. Feedback and nonfeedback systems

### 2.8.1 Static and Dynamic Systems

A continuous time system is called **static** or **memoryless** if its output at any instant of time  $t$  depends at most on the input signal at the same time but not on the past or future input. In any other case, the system is said to be **dynamic** or to have memory.

**Example :**

$$\begin{array}{lcl}
 y(t) = a x(t) & & \\
 y(t) = t x(t) + 6 x^3(t) & \left. \vphantom{\begin{array}{l} y(t) = a x(t) \\ y(t) = t x(t) + 6 x^3(t) \end{array}} \right\} & \text{Static systems} \\
 \\
 y(t) = t x(t) + 3 x(t^2) & & \\
 y(t) = x(t) + 3 x(t-2) & \left. \vphantom{\begin{array}{l} y(t) = t x(t) + 3 x(t^2) \\ y(t) = x(t) + 3 x(t-2) \end{array}} \right\} & \text{Dynamic systems}
 \end{array}$$

### 2.8.2 Time Invariant and Time Variant Systems

A system is said to be **time invariant** if its input-output characteristics does not change with time.

**Definition :** A relaxed system  $\mathcal{H}$  is **time invariant** or **shift invariant** if and only if

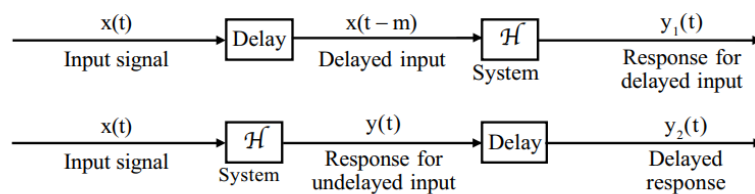
$$x(t) \xrightarrow{\mathcal{H}} y(t) \text{ implies that, } x(t-m) \xrightarrow{\mathcal{H}} y(t-m)$$

for every input signal  $x(t)$  and every time shift  $m$ .

i.e., in time invariant systems, if  $y(t) = \mathcal{H}\{x(t)\}$  then  $y(t-m) = \mathcal{H}\{x(t-m)\}$ .

A system  $\mathcal{H}$  is **time invariant** if the response to a shifted (or delayed) version of the input is identical to a shifted (or delayed) version of the response based on the unshifted (or undelayed) input.

The diagrammatic explanation of the above definition of time invariance is shown in fig 2.41.



If  $y_1(t) = y_2(t)$  then the system is time invariant

#### Procedure to test for time invariance

1. Delay the input signal by  $m$  units of time and determine the response of the system for this delayed input signal. Let this response be  $y_1(t)$ .
2. Delay the response of the system for unshifted input by  $m$  units of time. Let this delayed response be  $y_2(t)$ .
3. Check whether  $y_1(t) = y_2(t)$ . If they are equal then the system is time invariant. Otherwise the system is time variant.

#### Example 2.11

State whether the following systems are time invariant or not.

- a)  $y(t) = 2t x(t)$       b)  $y(t) = x(t) \sin 20\pi t$       c)  $y(t) = 3x(t^2)$       d)  $y(t) = x(-t)$

### 2.8.3 Linear and Nonlinear Systems

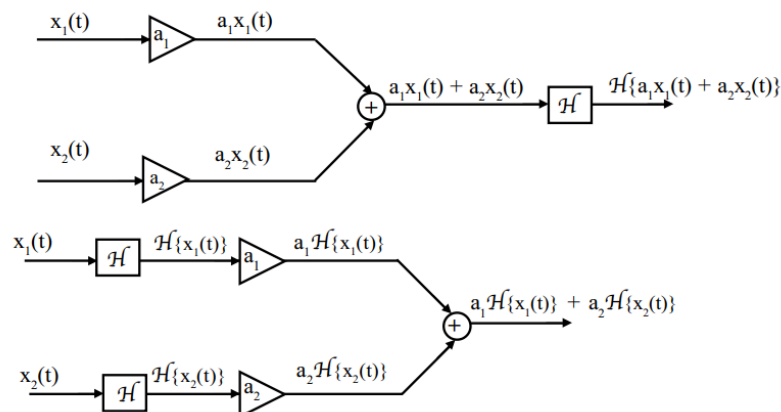
A **linear system** is the one that satisfies the superposition principle.

The **principle of superposition** requires that the response of a system to a weighted sum of the signals is equal to the corresponding weighted sum of the responses to each of the individual input signals.

**Definition :** A relaxed system  $\mathcal{H}$  is **linear** if

$$\mathcal{H}\{a_1 x_1(t) + a_2 x_2(t)\} = a_1 \mathcal{H}\{x_1(t)\} + a_2 \mathcal{H}\{x_2(t)\}$$

for any arbitrary input signal  $x_1(t)$  and  $x_2(t)$  and for any arbitrary constants  $a_1$  and  $a_2$ .



The system,  $\mathcal{H}$  is linear if and only if,  $\mathcal{H}\{a_1 x_1(t) + a_2 x_2(t)\} = a_1 \mathcal{H}\{x_1(t)\} + a_2 \mathcal{H}\{x_2(t)\}$

#### Procedure to test for linearity

1. Let  $x_1(t)$  and  $x_2(t)$  be two inputs to the system  $\mathcal{H}$ , and  $y_1(t)$  and  $y_2(t)$  be the corresponding responses.
2. Consider a signal,  $x_3(t) = a_1 x_1(t) + a_2 x_2(t)$  which is a weighed sum of  $x_1(t)$  and  $x_2(t)$ .
3. Let  $y_3(t)$  be the response for  $x_3(t)$ .
4. Check whether  $y_3(t) = a_1 y_1(t) + a_2 y_2(t)$ . If equal then the system is linear, otherwise it is nonlinear.

#### Example 2.13

Test the following systems for linearity.

- a)  $y(t) = t x(t)$ ,      b)  $y(t) = x(t^2)$ ,      c)  $y(t) = x^2(t)$ ,      d)  $y(t) = A x(t) + B$ ,      e)  $y(t) = e^{x(t)}$ .

### 2.8.4 Causal and Noncausal Systems

**Definition :** A system is said to be **causal** if the output of the system at any time  $t$  depends only on the present input, past inputs and past outputs but does not depend on the future inputs and outputs.

If the system output at any time  $t$  depends on future inputs or outputs then the system is called a **noncausal** system.

The causality refers to a system that is realizable in real time. It can be shown that an LTI system is causal if and only if the impulse response is zero for  $t < 0$ , (i.e.,  $h(t) = 0$  for  $t < 0$ ).

#### Example 2.15

Test the causality of the following systems.

$$a) y(t) = x(t) - x(t-1)$$

$$b) y(t) = x(t) + 2x(3-t)$$

$$c) y(t) = t x(t)$$

$$d) y(t) = x(t) + \int_0^t x(\lambda) d\lambda$$

$$e) y(t) = x(t) + \int_0^{3t} x(\lambda) d\lambda$$

$$f) y(t) = 2x(t) + \frac{dx(t)}{dt}$$

### 2.8.5 Stable and Unstable Systems

**Definition :** An arbitrary relaxed system is said to be **BIBO stable** (Bounded Input-Bounded Output stable) if and only if every bounded input produces a bounded output.

Let  $x(t)$  be the input of continuous time system and  $y(t)$  be the response or output for  $x(t)$ .

The term **bounded input** refers to finite value of the input signal  $x(t)$  for any value of  $t$ . Hence if input  $x(t)$  is bounded then there exists a constant  $M_x$  such that  $|x(t)| \leq M_x$  and  $M_x < \infty$ , for all  $t$ .

Examples of bounded input signal are step signal, decaying exponential signal and impulse signal.

Examples of unbounded input signal are ramp signal and increasing exponential signal.

The term **bounded output** refers to finite and predictable output for any value of  $t$ . Hence if output  $y(t)$  is bounded then there exists a constant  $M_y$  such that  $|y(t)| \leq M_y$  and  $M_y < \infty$ , for all  $t$ .

In general, the test for stability of the system is performed by applying specific input. On applying a bounded input to a system if the output is bounded then the system is said to be BIBO stable.

#### Condition for Stability of an LTI System

For an LTI (Linear Time Invariant) system, the condition for BIBO stability can be transformed to a condition on impulse response,  $h(t)$ . For BIBO stability of an LTI continuous time system, the integral of impulse response should be finite.

$$\therefore \int_{-\infty}^{+\infty} |h(t)| dt < \infty, \text{ for stability of an LTI system.}$$

#### Example 2.18

Test the stability of the LTI systems, whose impulse responses are given below.

$$a) h(t) = e^{-5|t|}$$

$$b) h(t) = e^{4t} u(t)$$

$$c) h(t) = e^{-4t} u(t)$$

$$d) h(t) = t e^{-3t} u(t)$$

$$e) h(t) = t \cos t u(t)$$

$$f) h(t) = e^{-t} \sin t u(t)$$

### 2.8.6 Feedback and Nonfeedback Systems

The system in which the output  $y(t)$  at any time  $t$  depends on past output, past input and present input is called a **feedback system**. The integration and differentiation of a signal at any time depends on past value and so the equations governing feedback systems will have terms involving differentiations and integrations of output and input.

The equations governing feedback systems will be in the form,

$$a_0 \frac{d^N}{dt^N} y(t) + a_1 \frac{d^{N-1}}{dt^{N-1}} y(t) + a_2 \frac{d^{N-2}}{dt^{N-2}} y(t) + \dots + a_{N-1} \frac{d}{dt} y(t) + a_N y(t) = b_0 \frac{d^M}{dt^M} x(t) + b_1 \frac{d^{M-1}}{dt^{M-1}} x(t) + b_2 \frac{d^{M-2}}{dt^{M-2}} x(t) + \dots + b_{M-1} \frac{d}{dt} x(t) + b_M x(t)$$

The system in which the output depends only on the present and past input is called a **nonfeedback system**. The equations governing nonfeedback systems will not have terms involving differentiations and integrations of output.

The equations governing nonfeedback systems will be in the form,

$$y(t) = b_0 \frac{d^M}{dt^M} x(t) + b_1 \frac{d^{M-1}}{dt^{M-1}} x(t) + b_2 \frac{d^{M-2}}{dt^{M-2}} x(t) + \dots + b_{M-1} \frac{d}{dt} x(t) + b_M x(t)$$

## 2.9 Convolution of Continuous Time Signals

The **convolution** of two continuous time signals  $x_1(t)$  and  $x_2(t)$  is defined as,

$$x_3(t) = \int_{-\infty}^{+\infty} x_1(\lambda) x_2(t - \lambda) d\lambda \quad \dots(2.21)$$

where,  $x_3(t)$  is the signal obtained by convolving  $x_1(t)$  and  $x_2(t)$ , and  $\lambda$  is a dummy variable used for integration.

The convolution relation of equation (2.21) can be symbolically expressed as,

$$x_3(t) = x_1(t) * x_2(t) \quad \dots (2.22)$$

where the symbol  $*$  indicates convolution operation.

### 2.9.2 Properties of Convolution

The convolution of continuous time signals will satisfy the following properties.

**Commutative property** :  $x_1(t) * x_2(t) = x_2(t) * x_1(t)$

**Associative property** :  $[x_1(t) * x_2(t)] * x_3(t) = x_1(t) * [x_2(t) * x_3(t)]$

**Distributive property** :  $x_1(t) * [x_2(t) + x_3(t)] = [x_1(t) * x_2(t)] + [x_1(t) * x_3(t)]$

### 2.9.4 Procedure to Perform Convolution

The **convolution** of two continuous time signals  $x_1(t)$  and  $x_2(t)$  is defined as,

$$x_3(t) = x_1(t) * x_2(t) = \int_{-\infty}^{+\infty} x_1(\lambda) x_2(t - \lambda) d\lambda$$

where,  $x_3(t)$  is the signal obtained by convolving  $x_1(t)$  and  $x_2(t)$ ,  
 $\lambda$  is a dummy variable used for integration,  
 $*$  indicates convolution operation.

The computation of  $x_3(t)$  using the above convolution equation for any value of  $t$  involves the following operations,

1. **Change of time index** : The time index  $t$  in signals  $x_1(t)$  and  $x_2(t)$  is changed to  $\lambda$  to get  $x_1(\lambda)$  and  $x_2(\lambda)$ .
2. **Folding** : The signal  $x_2(\lambda)$  is folded to get  $x_2(-\lambda)$ .
3. **Shifting** : The signal  $x_2(-\lambda)$  is shifted by  $t$  units of time to get  $x_2(t-\lambda)$ .
4. **Multiplication** : The signals  $x_1(\lambda)$  and  $x_2(t-\lambda)$  are multiplied to get a product signal.
5. **Integration** : The product signal is integrated to get  $x_3(t)$ . Let the product signal is nonzero in the interval  $\lambda=\lambda_1$  to  $\lambda=\lambda_2$ , Now the signal  $x_3(t)$  is given by,

$$x_3(t) = \int_{\lambda=\lambda_1}^{\lambda=\lambda_2} x_1(\lambda) x_2(t - \lambda) d\lambda$$

If both  $x_1(t)$  and  $x_2(t)$  are defined for  $t > 0$ , (i.e., both  $x_1(t)$  and  $x_2(t)$  are causal) then the product signal is nonzero in the interval  $\lambda=0$  to  $\lambda=t$ , Now the signal  $x_3(t)$  is given by,

$$x_3(t) = \int_{\lambda=0}^{\lambda=t} x_1(\lambda) x_2(t - \lambda) d\lambda$$

### Example 2.23

Perform convolution of the following signals, by graphical method and sketch the resultant signal.

